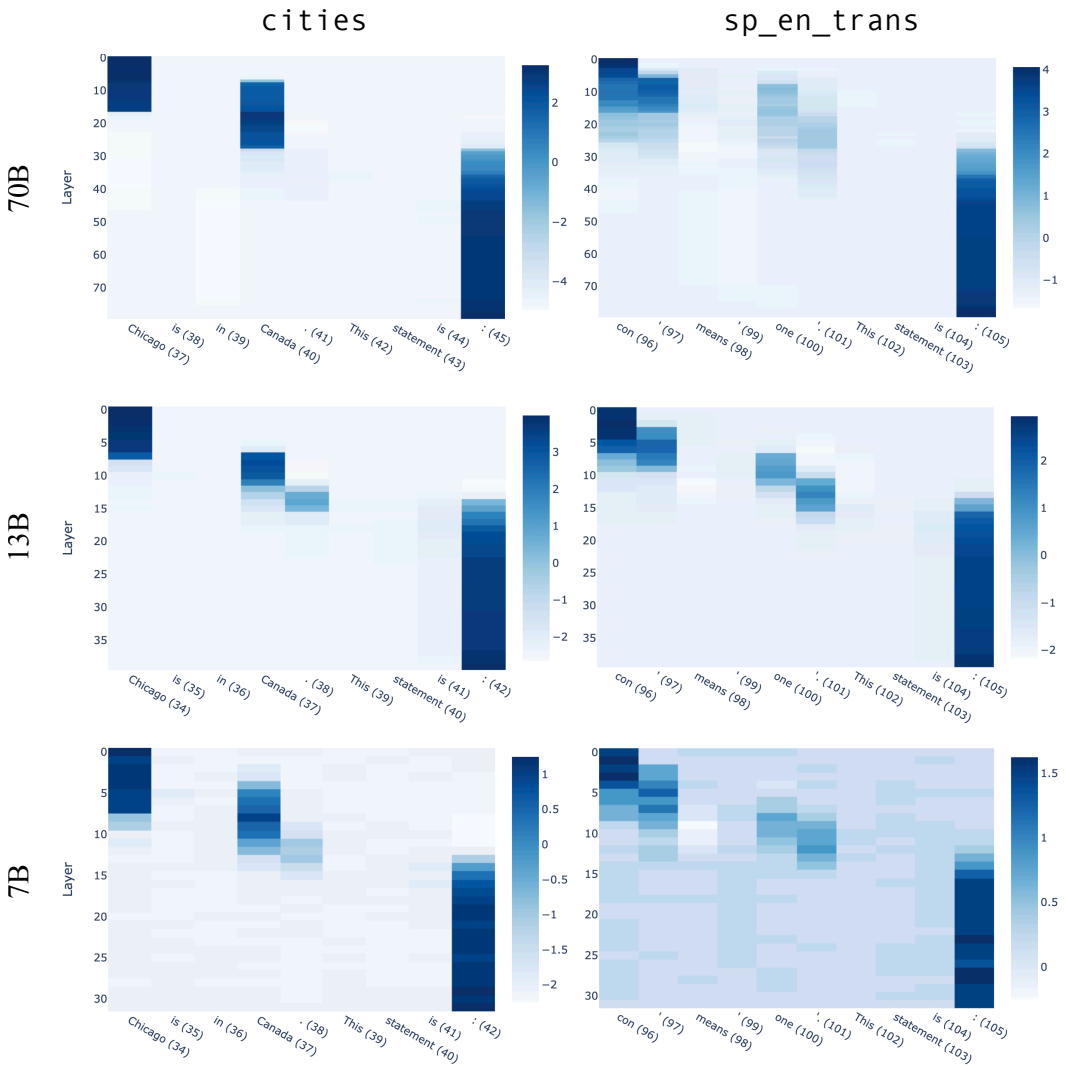

Figure 6: Full patching results across all three model sizes and inputs. Results are for patching false inputs (shown) to true by changing the first token shown on the left. Numbers in parentheses are the index of the token in the full (few-shot) prompt.

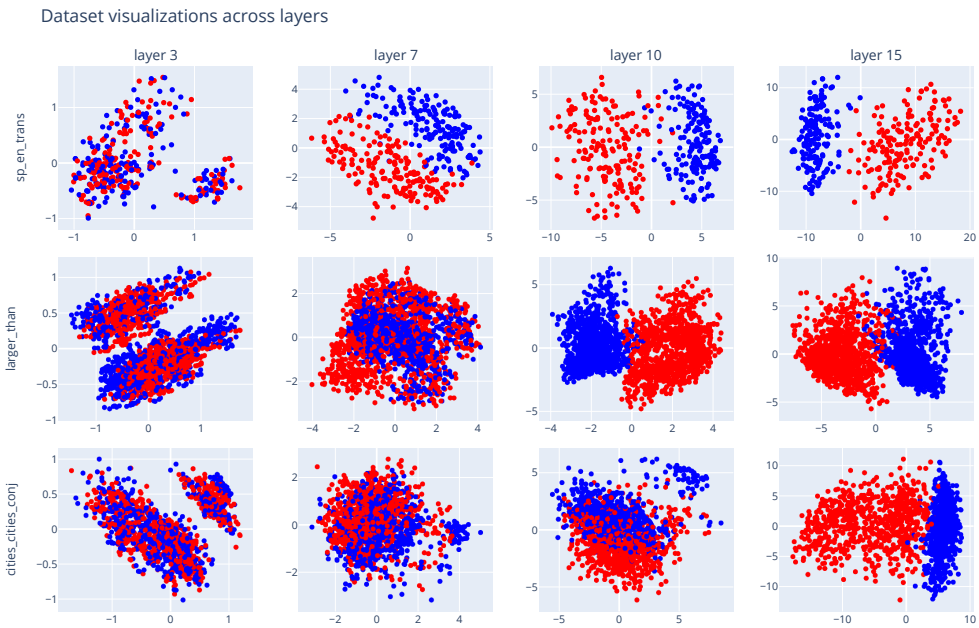

Figure 7: Projections of LLaMA-2-13B representations of datasets onto their top two PCs, across various layers.

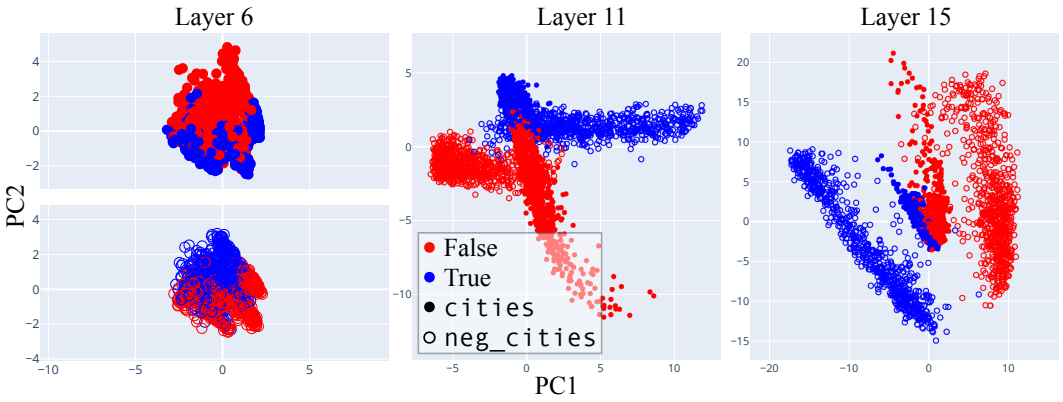

Figure 8: PCA visualizations of LLaMA-2-13B representations of `cities` and `neg_cities` at various layers.

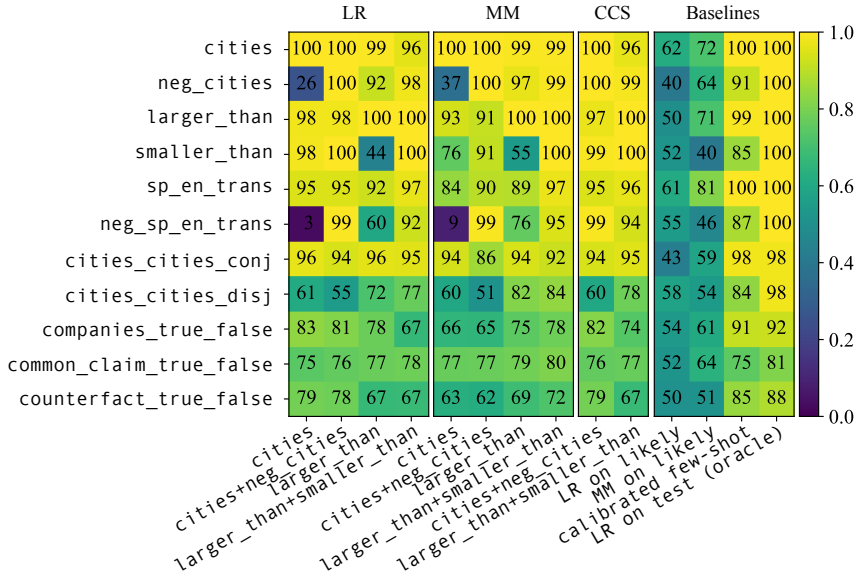

Figure 9: Generalization results for **LLaMA-2-70B**.

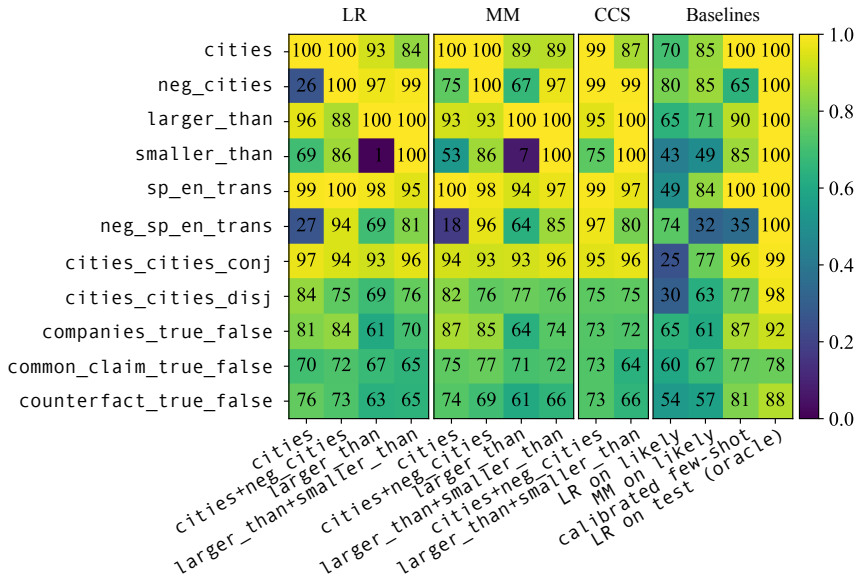

Figure 10: Generalization results for **LLaMA-2-13B**.

D Full generalization results

Here we present the full generalization results for probes trained on LLaMA-2-70B (Fig. 9), 13B (Fig. 10), and 7B (Fig. 11). The horizontal axis shows the training data for the probe and the vertical axis shows the test set.

E Mass-mean probing in terms of Mahalanobis whitening

One way to interpret the formula $p^{\text{iid}}_{\text{mm}}(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}_{\text{mm}}^T \Sigma^{-1} \boldsymbol{x})$ for the IID version of mass-mean probing is in terms of Mahalanobis whitening. Recall that if $\mathcal{D} = \{x_i\}$ is a dataset of $x_i \in \mathbb{R}^d$ with covariance matrix Σ, then the Mahalanobis whitening transformation $W = \Sigma^{-1/2}$

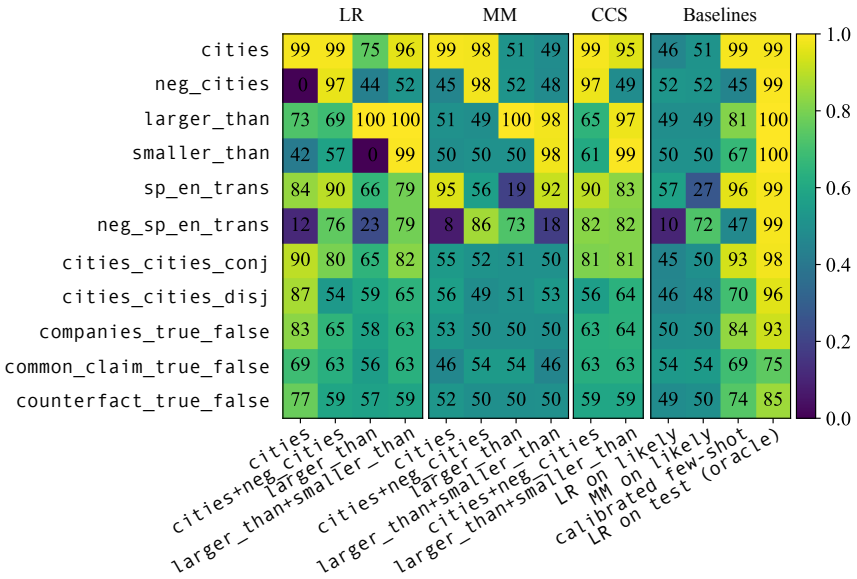

Figure 11: Generalization results for **LLaMA-2-7B**.

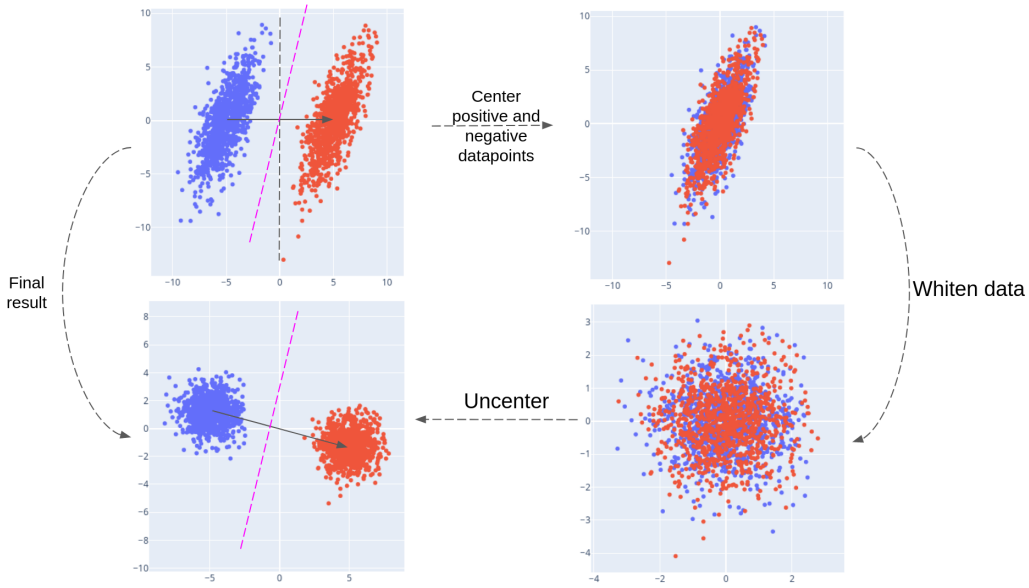

Figure 12: Mass-mean probing is equivalent to taking the projection onto $\boldsymbol{\theta}_{\mathrm{mm}}$ after applying a whitening transformation.

satisfies the property that $\mathcal{D}' = \{Wx_i\}$ has covariance matrix given by the identity matrix, i.e. the whitened coordinates are uncorrelated with variance 1. Thus, noting that $\boldsymbol{\theta}_{\mathrm{mm}}^T \Sigma^{-1} \boldsymbol{x}$ coincides with the inner product between $W\boldsymbol{x}$ and $W\boldsymbol{\theta}$, we see that p_{mm} amounts to taking the projection onto $\boldsymbol{\theta}_{\mathrm{mm}}$ after performing the change-of-basis given by W. This is illustrated with hypothetical data in Fig. 12.

F For Gaussian data, IID mass-mean probing coincides with logistic regression on average

Let $\boldsymbol{\theta} \in \mathbb{R}^d$ and Σ be a symmetric, positive-definite $d \times d$ matrix. Suppose given access to a distribution $\mathcal{D}$ of datapoints $\boldsymbol{x} \in \mathrm{R}^d$ with binary labels $y \in \{0, 1\}$ such that the negative datapoints are distributed as $\mathcal{N}(-\theta, \Sigma)$ and the positive datapoints are distributed as $\mathcal{N}(\theta, \Sigma)$. Then the vector identified by mass-mean probing is $\boldsymbol{\theta}_{\mathrm{mm}} = 2\boldsymbol{\theta}$. The following theorem then shows that $p_{\mathrm{mm}}^{\mathrm{iid}}(\boldsymbol{x}) = \sigma(2\theta^T \Sigma^{-1} \boldsymbol{x})$ is also the solution to logistic regression up to scaling.

Theorem F.1. *Let*

$$\boldsymbol{\theta}_{\mathrm{lr}} = \underset{\phi : \|\phi\| = 1}{\arg\min}\, \mathbb{E}_{(\boldsymbol{x}, y) \sim \mathcal{D}} \left[y \log \sigma\left(\phi^T \boldsymbol{x}\right) + (1 - y) \log\left(1 - \sigma\left(\phi^T \boldsymbol{x}\right)\right) \right]$$

be the direction identified by logistic regression. Then $\boldsymbol{\theta}_{\mathrm{lr}} \propto \Sigma^{-1} \boldsymbol{\theta}$.

Proof. Since the change of coordinates $\boldsymbol{x} \mapsto W\boldsymbol{x}$ where $W = \Sigma^{-1/2}$ (see App. E) sends $\mathcal{N}(\pm\theta, \Sigma)$ to $\mathcal{N}(\pm W\theta, I_d)$, we see that

$$W\Sigma\boldsymbol{\theta}_{\mathrm{lr}} = \underset{\phi : \|\phi\| = 1}{\arg\min}\, \mathbb{E}_{(\boldsymbol{x}, y) \sim \mathcal{D}'} \left[y \log \sigma\left(\phi^T W\boldsymbol{x}\right) + (1 - y) \log\left(1 - \sigma\left(\phi^T W\boldsymbol{x}\right)\right) \right]$$

where $\mathcal{D}'$ is the distribution of labeled $\boldsymbol{x} \in \mathbb{R}^d$ such that the positive/negative datapoints are distributed as $\mathcal{N}(\pm W\theta, I_d)$. But the argmax on the right-hand side is clearly $\propto W\theta$, so that $\theta_{\mathrm{lr}} \propto \Sigma^{-1}\theta$ as desired. □

G Difference-in-means directions and linear concept erasure

In this appendix, we explain the connection between difference-in-means directions and optimal erasure. One consequence of this connection is that it suggests a natural extension of difference-in-means probes to multi-class classification data.

The connection comes via the following theorem from Belrose et al. (2023).

Theorem G.1. *(Belrose et al., 2023, Thm. 3.1.) Let* (X, Y) *be jointly distributed random vectors with* $X \in \mathbb{R}^d$ *having finite mean and* $Y \in \mathcal{Y} = \{\mathbf{y} \in \{0, 1\}^k : \|\mathbf{y}\|_1 = 1\}$ *(representing one-hot encodings of a multi-class labels). Suppose that* $\mathcal{L} : \mathbb{R}^k \times \mathcal{Y} \to \mathbb{R}^{>0}$ *is a loss function convex in its first argument (e.g. cross-entropy loss).*

If the class-conditional means $\mathbb{E}[X|Y = i]$ *for* $i \in \{1, \ldots, k\}$ *are all equal, then the best affine predictor (that is, a predictor* $\eta : \mathbb{R}^d \to \mathbb{R}^k$ *of the form* $\eta(\boldsymbol{x}) = W\boldsymbol{x} + \boldsymbol{b}$*) is constant* $\eta(\boldsymbol{x}) = \boldsymbol{b}$.

In the case of a binary classification problem (X, Y), this theorem implies that any nullity 1 projection P which eliminates linearly-recoverable information from X has kernel

$$\ker P = \operatorname{span}(\boldsymbol{\delta})$$

generated by the difference-in-mean vector $\boldsymbol{\delta} = \boldsymbol{\mu}^+ - \boldsymbol{\mu}^-$ for the classes.

For a more general multi-class classification problem, one could similarly ask: What is the "best" direction to project away in order to eliminate linearly-recoverable information from X? A natural choice is thus the top left singular vector of the cross-covariance matrix Σ_{XY}. (In the case of binary classification, we have that $\Sigma_{XY} = [-\boldsymbol{\delta} \;\; \boldsymbol{\delta}]$ has column rank 1, making $\boldsymbol{\delta}$ the top left singular vector.)